

3D Human Pose Estimation Using Ultra-low Resolution Thermal Images

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Abstract

Can we estimate 3D human pose from ultra-low resolution thermal images (e.g., 8×8 pixels)? This study explores this possibility. Thermal images capture radiation intensity, minimizing personal information exposure, and are commonly used in devices like air conditioners. We propose a framework that uses 8×8 thermal images for 3D human pose estimation, enhancing privacy and efficiency. To overcome challenges from subject and ambient temperature variations, we employ adversarial learning with discriminators for subject and temperature, ensuring robust and invariant feature extraction.

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1 Introduction

3D human pose estimation (HPE) has various applications (e.g., sports analysis, surveillance). Existing methods using various modalities, such as RGB images, face challenges in low-light conditions, privacy concerns, and susceptibility to interference. Infrared sensors capturing thermal images provide a solution by preserving privacy and are less prone to interference from electronic devices.

This paper proposes a novel framework for 3D HPE using ultra-low resolution (8×8 pixels) thermal images. However, this approach presents unique challenges, including low resolution, susceptibility to subject and temperature variations, and the absence of existing datasets.

To address these challenges, our technical contributions are as follows: (1) we develop an end-to-end data processing pipeline that directly maps low-dimensional thermal images to human poses, and (2) introduce adversarial learning with discriminators for subject and ambient temperature in-variance. Additionally, (3)

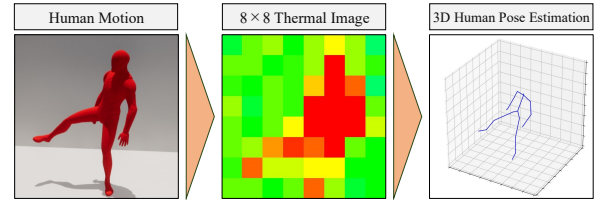


Figure 1: 3D HPE from 8×8 pixel thermal images captured by a single infrared sensor.

we generate synthetic data to improve estimation accuracy and facilitate future dataset creation.

2 Method

Thermal Image-Based HPE Framework. Our proposed framework consists of the HPE model f , a subject discriminator module D_{subj} , and an ambient temperature discriminator module D_{temp} . These are introduced using adversarial learning to enhance robustness against variations in subjects and ambient temperatures. θ_f , containing all trainable parameters for f , is updated based on the gradients computed from the following loss function (1), defined as a linear combination of the MSE loss of the 3D pose in f , the cross-entropy losses of the two modules, and weight parameters w_α , w_β , and w_γ :

$$\mathcal{L}_{\text{total}}(\theta_f) = w_\alpha \mathcal{L}_{\text{pose}} + w_\beta \mathcal{L}_{\text{subj}} + w_\gamma \mathcal{L}_{\text{temp}}. \quad (1)$$

Data Synthesis. Since our approach addresses a novel task and there is a lack of sufficient datasets, we employ data synthesis. For

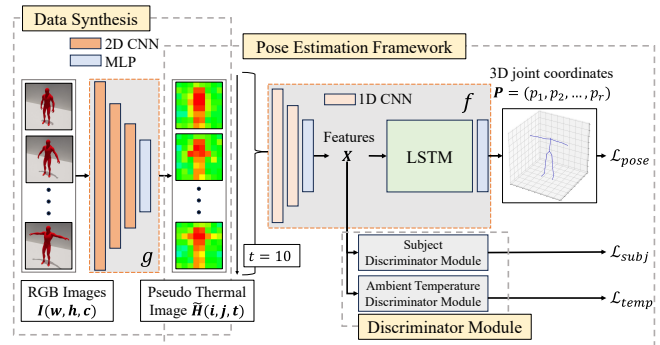


Figure 2: Overview of the proposed method

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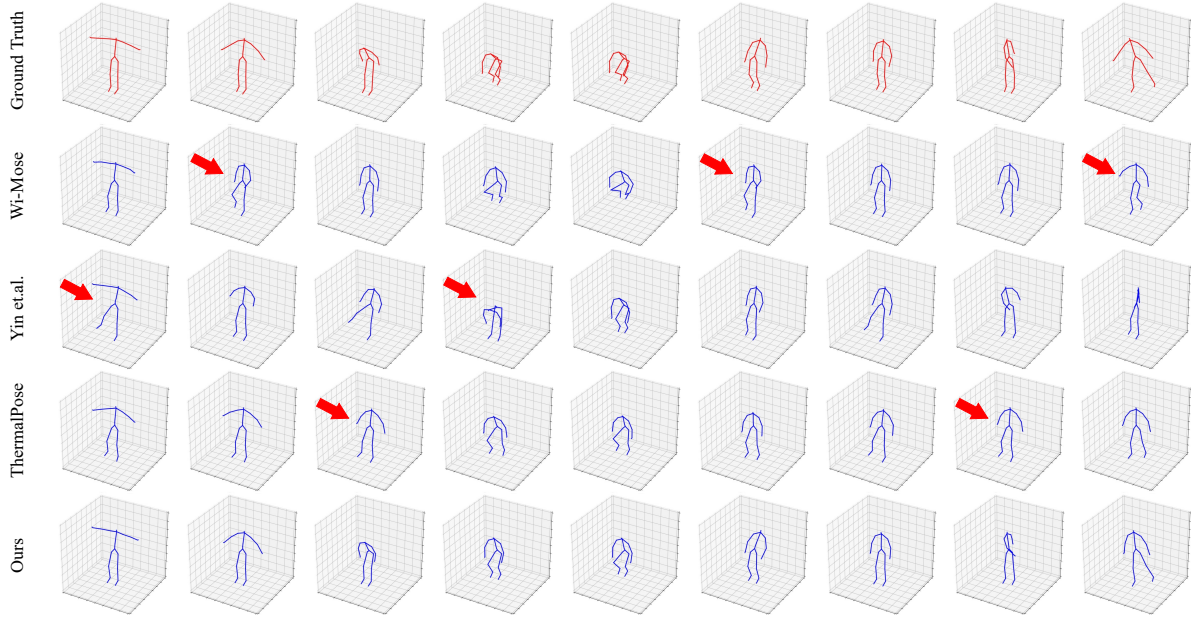


Figure 3: Qualitative results of the proposed method compared to ground truth.

data synthesis, we place avatars in a virtual environment (Unreal Engine) and replay motion data to render RGB images. These images are then fed into a pseudo-thermal image generation model g , which produces 8×8 pseudo-thermal images. A noteworthy aspect of this method is that it allows us to generate thermal images corresponding to any arbitrary motion data.

Experimental Setup. All real-world data were measured indoors using a Grid-EYE AMG8833 infrared sensor. Ground truth data for human poses were obtained using an OptiTrack motion capture system. We collected data from 10 male subjects, with three different temperature settings per subject. Half the data was used to generate synthetic data, while the remaining was used to test pose estimation. Evaluation metrics include Mean Per Joint Position Error (MPJPE) and Percentage of Correct Keypoints (PCK), with the PCK threshold set at 50% of the head-to-neck distance.

3 Results and Discussion

Quantitative Evaluation. We tested our method using real-world datasets. To ensure rigorous evaluation under novel conditions, we used data from subjects and ambient temperatures not included in the synthetic training dataset. Baseline methods (colored in yellow) such as Wi-Mose [Wang et al. 2021], which uses Wi-Fi signals for 3D HPE, the 8×8 thermal image-based pose classification [Yin et al. 2023], and ThermalPose [Chen et al. 2020], which performs 2D HPE using thermal images, were tested on the same datasets, but with their original architectures. Our approach (colored in green) demonstrated superior performance, as seen in Table 1. Furthermore, the model trained on 24K synthetic data (colored in blue) demonstrated higher accuracy than the model trained on the same amount of real data (colored in red).

Qualitative Analysis. The proposed method qualitatively shows more accurate pose estimation compared to baseline methods, as seen in Figure 3.

Table 1: Comparison between our method and baselines using different datasets. Baseline methods are evaluated with Synthetic Data 80K. RD: Real Data, SD: Synthetic Data.

Method	Trained on	MPJPE [mm] ↓	PCKh@0.5 ↑
Wi-Mose	SD 80K	140.43	0.60
Yin et al.	SD 80K	143.26	0.62
ThermalPose	SD 80K	129.82	0.60
Ours	SD 80K	70.90	0.86
Ours	SD 24K	79.02	0.83
Ours	RD 24K	83.69	0.80

4 Conclusion and Future Work

We introduced a novel method that successfully performs 3D HPE using 8×8 ultra-low resolution thermal images, demonstrating its effectiveness for this task. Future work will focus on improving the method’s adaptability by addressing the relationship between the sensor and the subject’s position. We will also explore real-time applications to enhance the practical usability of our method.

Acknowledgments

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